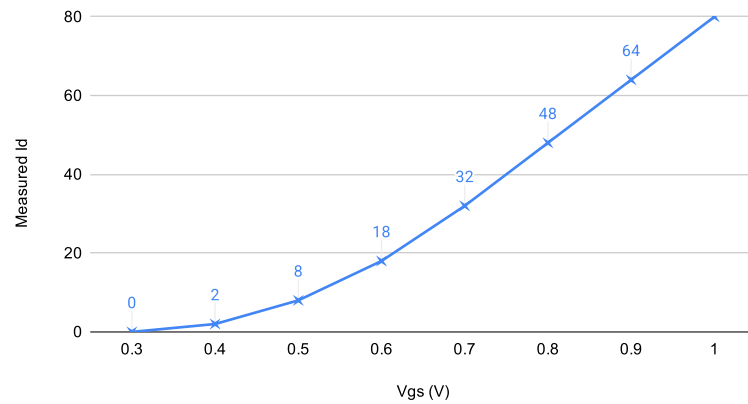


Vds (V) 1

Vgs (V)	Measured Id
0.3	0
0.4	2
0.5	8
0.6	18
0.7	32
0.8	48
0.9	64
1	80

Measured Id vs. Vgs (V)



Estimated Vt (V): 0.3

$$I_d = K(V_{gs} - V_t)^2$$
$$K = I_d / ((V_{gs} - V_t)^2)$$

Vgs	Estimated Vt	Id	K	Bold=Calculated
0.6	0.3	18	200	
1	0.3	98	200	< using calculated K
1	0.3	80	163.2653061	< Actual values

Computed K: 200
Predicted Id at Vgs=1: 98
Actual Id at Vgs=1: 80

QUADRATIC MODEL

Prediction Id is higher than measured value, and Prediction K is also higher than measured value.

Estimated Vt (V): 0.3

$$I_d = K_v(V_{gs} - V_t)$$
$$K_v = I_d / (V_{gs} - V_t)$$

Vgs	Estimated Vt	Id	K	Bold=Calculated
0.8	0.3	48	96	
1	0.3	67.2	96	< using calculated K
1	0.3	80	114.2857143	< Actual values

Computed K: 96
Predicted Id at Vgs=1: 67.2
Actual Id at Vgs=1: 80

LINEAR MODEL

Prediction Id is lower than measured value, and Prediction K is also lower than measured value.

Quadratic model:

Prediction Id is higher than measured value, and Prediction K is also higher than measured value.

Linear model:

Prediction Id is lower than measured value, and Prediction K is also lower than measured value.

	Quadratic Model:								
	$I_d = K(V_{gs} - V_t)^2$								
	$K = I_d / ((V_{gs} - V_t)^2)$								
	Prediction Id is higher than measured value, and Prediction K is also higher than measured value.								